Assignment: SEC210130 - Internetworking
Continuous Assignment 02

Total Marks: 100

Groups arrangement:

You need to arrange your own groups consisting of **2 to 3 members**. Please add the group names in to the LMS.

Assignment Scenario:

Suppose that the LNBTI Main Building does not have a proper computer network (LAN), and you have been assigned the task of updating the LNBTI Network. Conduct background research and collect the necessary information to design and propose a strong network for the building.

Assignment Questions:

Question 01: Equipment List (30 Marks)

- Prepare a detailed list of necessary equipment for each room on each floor of the LNBTI Main Building.
- Specify the PC requirements for each room.
- Include any additional devices such as printers, scanners, server computers, and other essential equipment.

Question 02: Cost Estimation (20 Marks)

- Create a cost estimation table for the proposed Local Area Network (LAN) for the LNBTI Main Building.
- Ensure that all costs, including devices, cabling, and installation charges and student and staff training sessions, are included.

Question 03: Network Design (50 Marks)

- Design a complete network diagram for the LNBTI Main Building, including correct notations and IP addresses for each floor. (Note: It is not necessary to draw all PCs; instead, mention the number of PCs available in each room.)
- Use subnetting methods to assign IP addresses and document all IP ranges and other relevant details for each room.

Submission Instructions:

- Demonstrate your propose network Diagram using software (cisco packet tracer).
- Make single proposal document and include answers to all the above questions.
- Submit your proposal in PDF format to the LMS before the submission deadline.

Assessment Criteria:

- **Completeness and accuracy** of the equipment list. (30 marks)
- **Feasibility and detail** in the cost estimation table. (20 marks)
- **Clarity and correctness** of the network diagram, including proper notations and IP addressing, Demonstration. (50 marks)